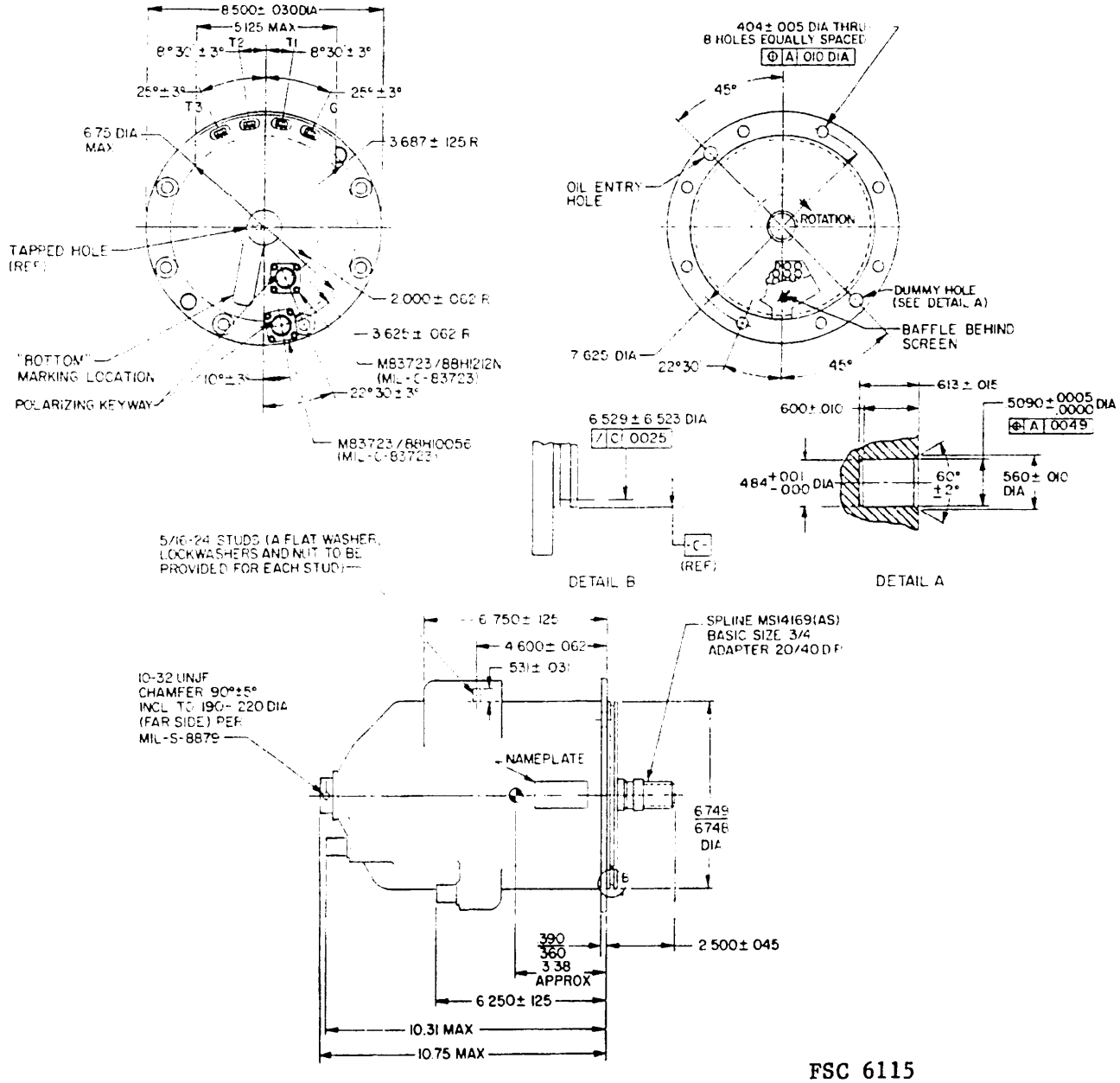
MILITARY SPECIFICATION SHEET

Generating System, Aircraft, 400 Hertz
3-Phase 115/200 Volt, 30/45 KVA

This specification is approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the system described herein shall consist of this document and the issue in effect of MIL-G-21480.



FSC 6115

FIGURE 1. Generator P/N M21480/23-2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

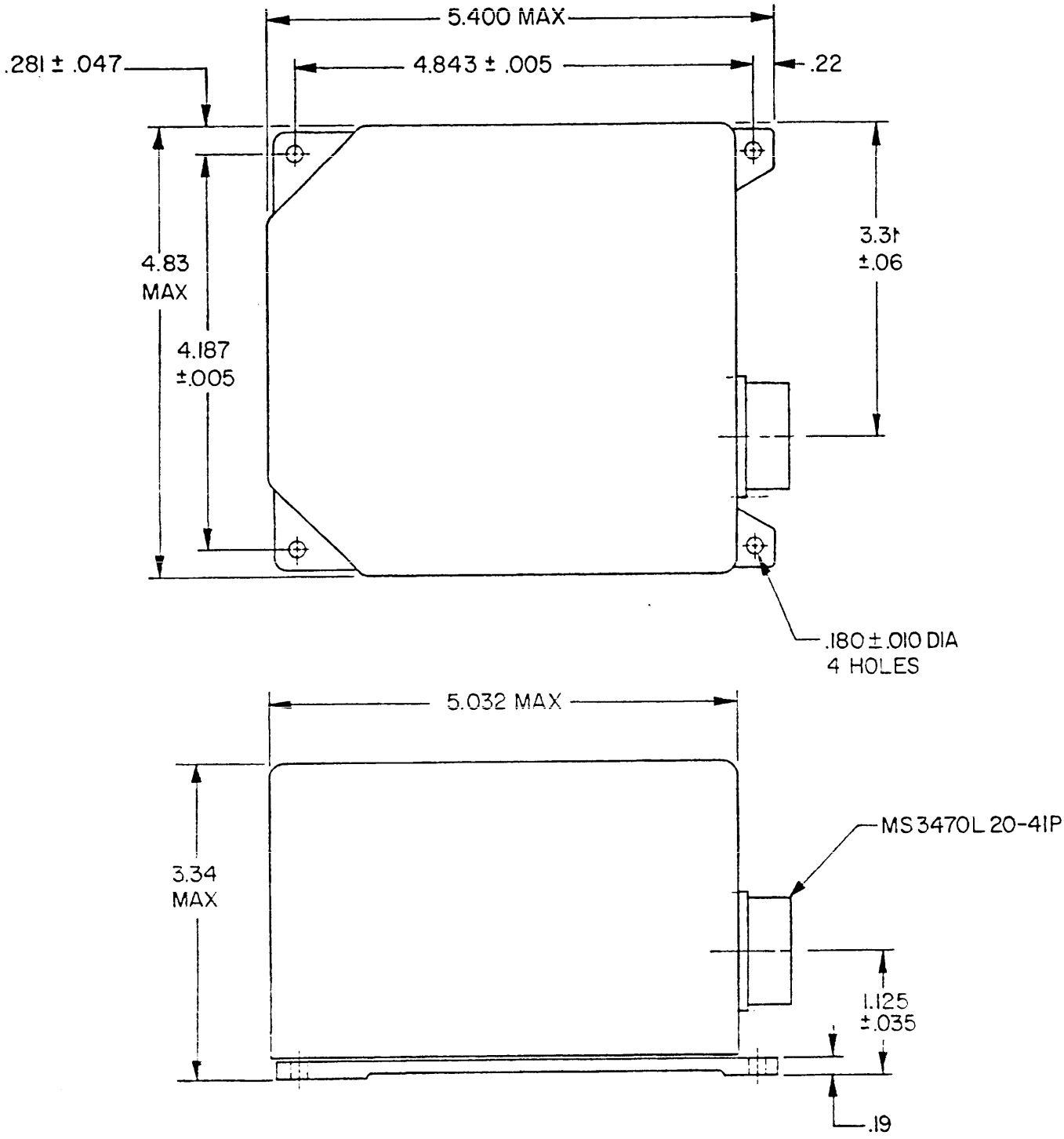


FIGURE 2. Generator control unit P/N M21480/23-3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

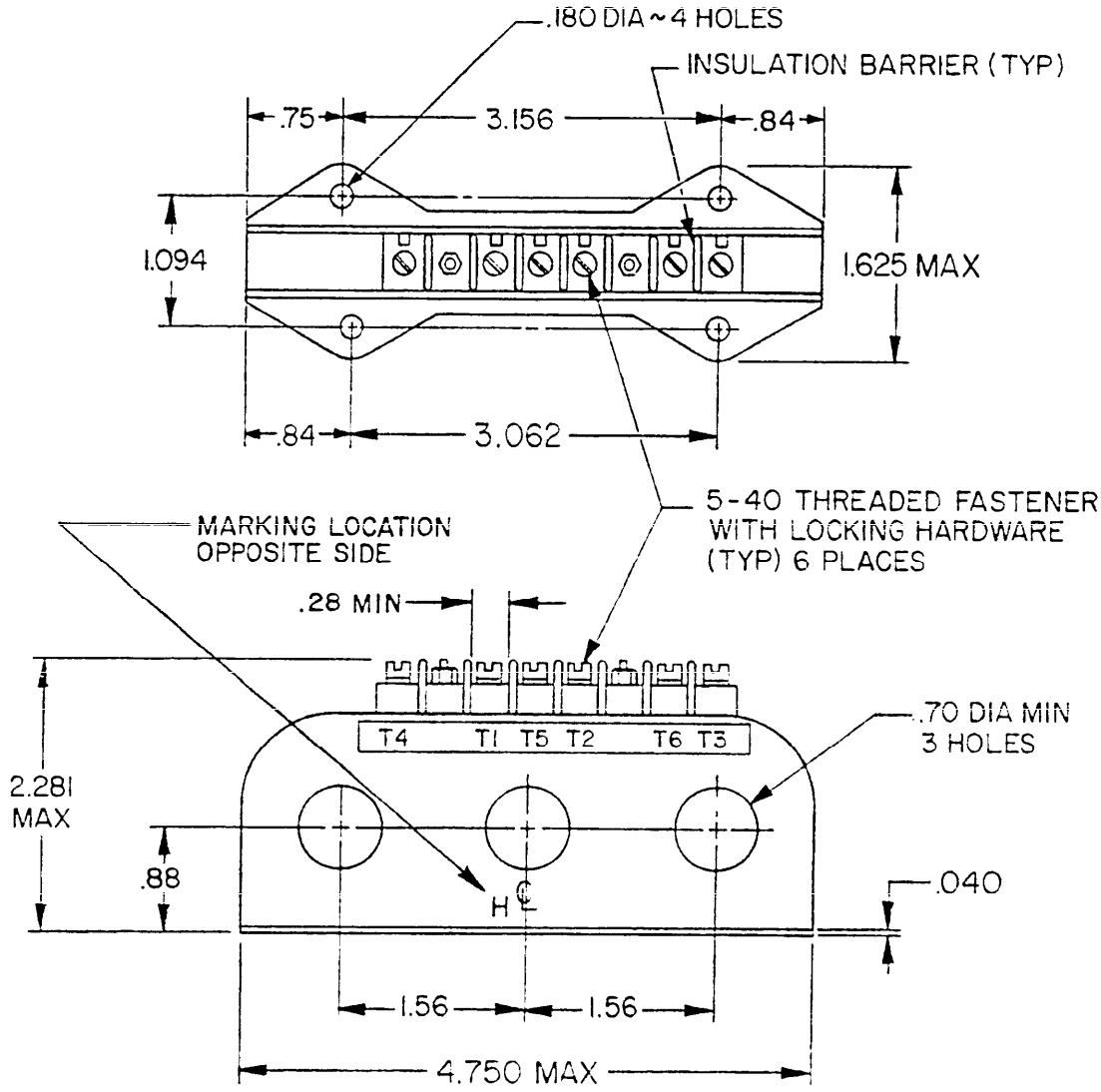


FIGURE 3. Current transformer P/N M21480/23-4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

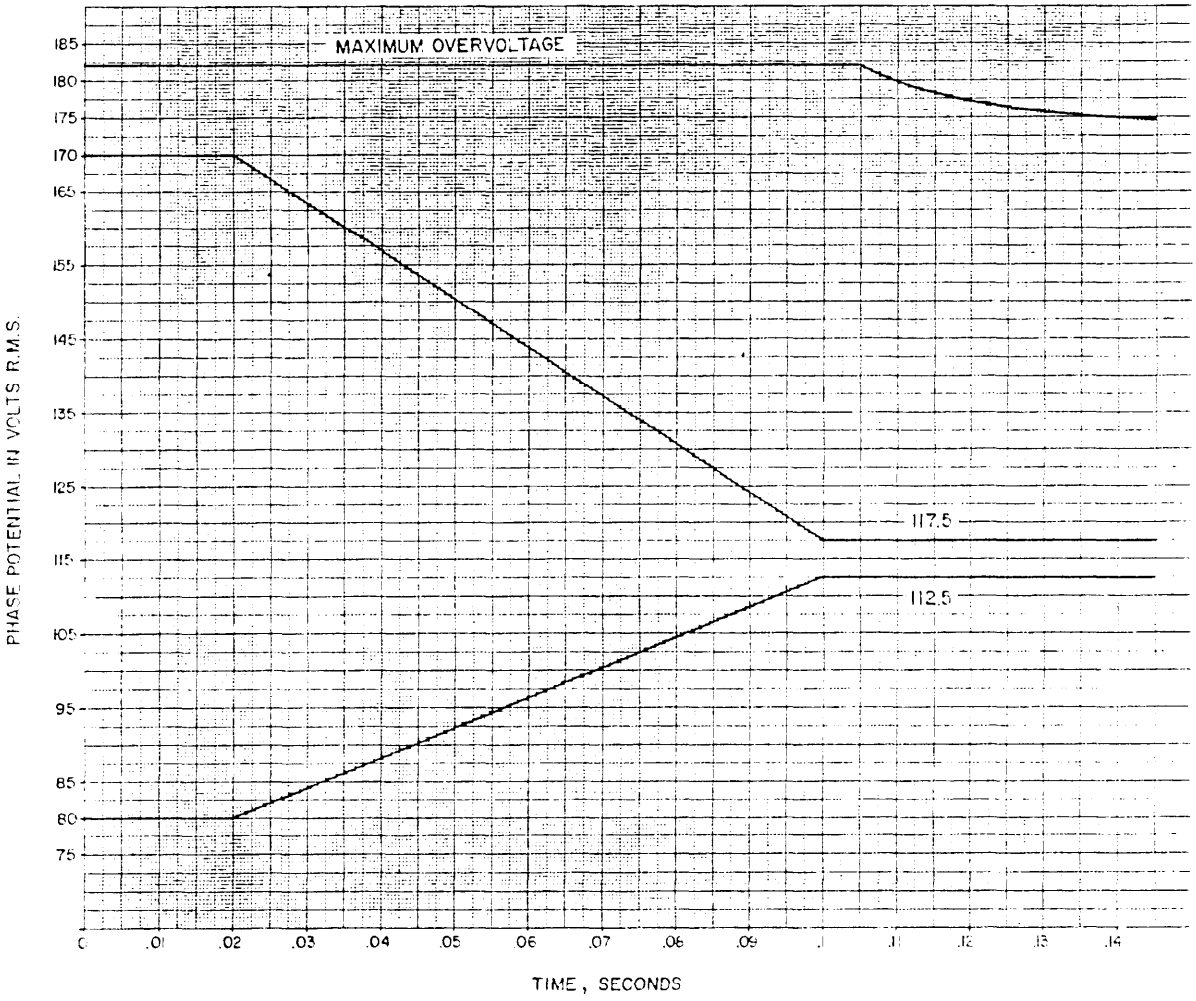


FIGURE 4. Transient limits and recovery time for application and removal of any load from zero to 200% rated load.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:44Z
Check the source to verify that this is the current version before use.

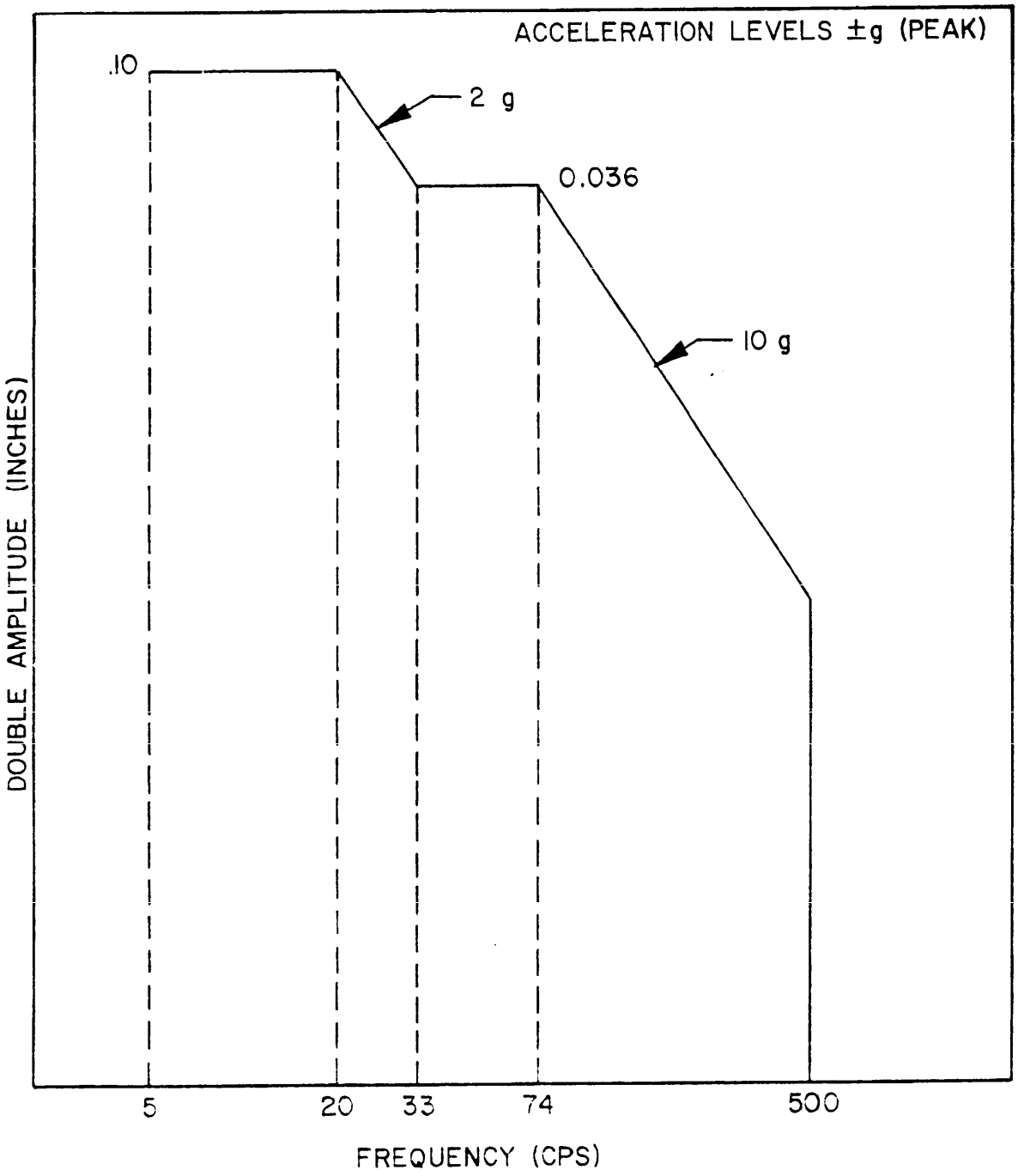


FIGURE 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

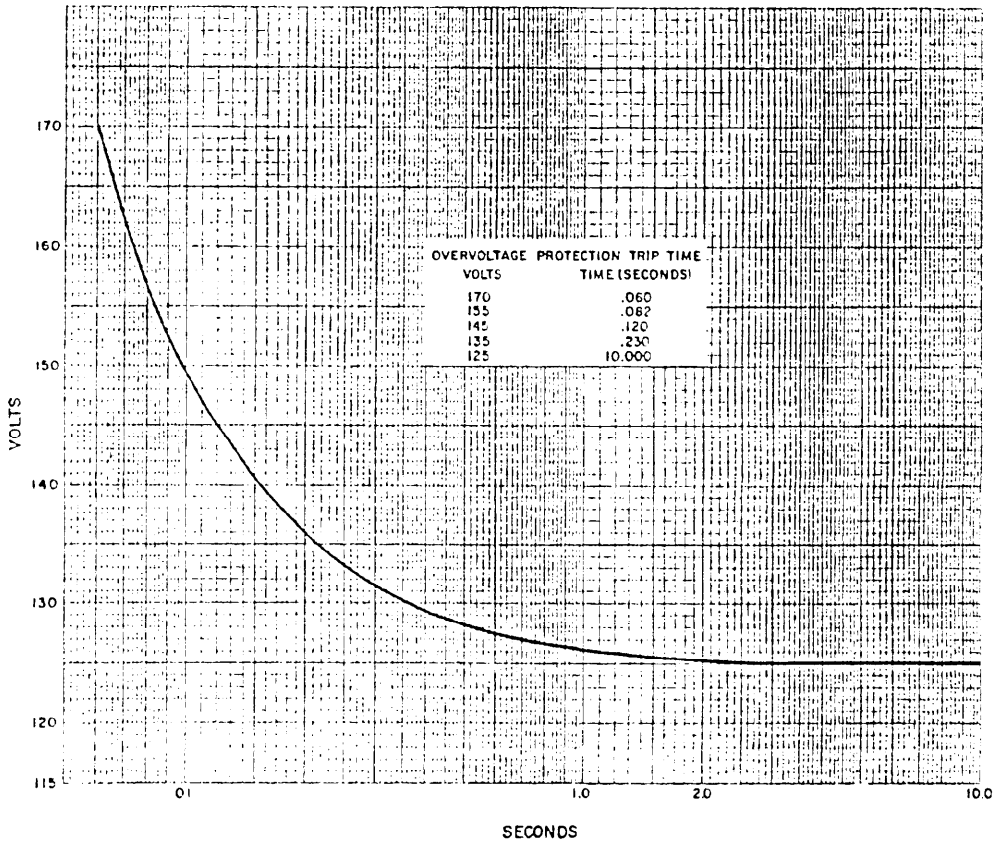
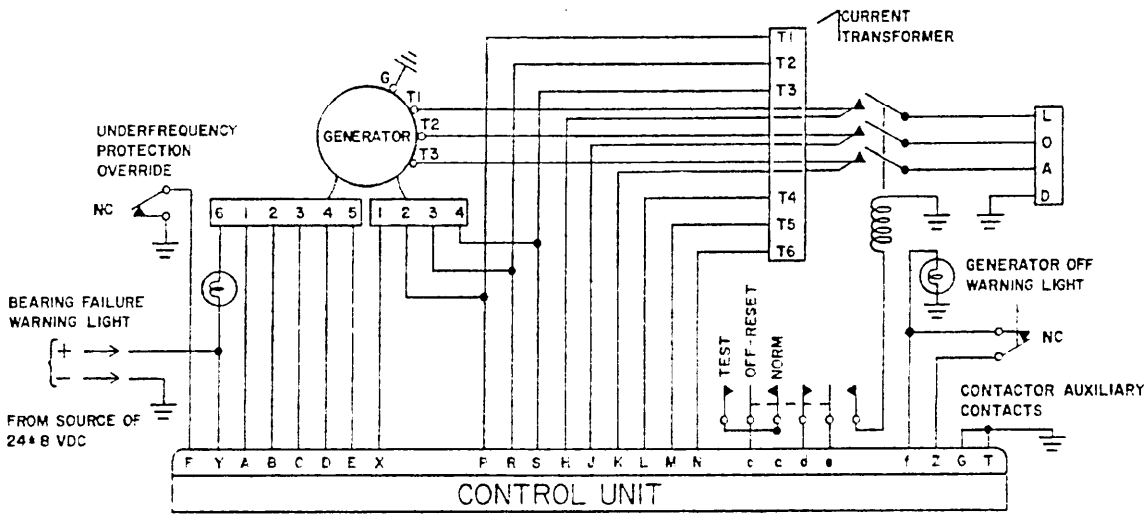


FIGURE 6. System ceiling voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.



Notes:

1. Tolerances. Unless otherwise specified, linear tolerances are: two-place decimals, ± .03; three-place decimals, ± .010. Angular ± 30°.

2. Components. Items which are shown on the test circuit but are not listed as part of the system on page 8 are not furnished under this specification sheet.

FIGURE 7. Test circuit P/N M21480/23-3.

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:44Z
Check the source to verify that this is the current version before use.

ITEM	MILITARY PART NUMBER
System Consists of: One M21480/23-2 Generator One M21480/23-3 Control Unit One M21480/23-4 Current Transformer	M21480/23-1
Generator	M21480/23-2
Control Unit	M21480/23-3
Current Transformer	M21480/23-4

FIGURE 8. Identification of items.

Generator	
Coolant Delivery	
Volume per minute	1 ± 0.03 gallons
Input Pressure Range	29.5 to 33.5 psi
Max. Oil Inlet Temp.	85°C
Operating Attitude	
Inclination of free end	18° down to 90° up
Rotation	30° CW to 30° CCW
	simultaneously with
	inclination of free
	end from zero to 18°
	down
Rated Speed	12,000 ± 600 RPM
Shear Section Strength	1,490 ± 179 in.-lb
Cooling Oil	Liquid
Above -40°C	MIL-L-23699
Below - 40°C	MIL-L-7808
Maximum Overhung Moment	98 in.-lb
Maximum Altitude	30,000 feet
Rated Load	45 kva at 0.75 lagging
	to unity power factor
Minimum Mean Time Between Failures	
System	2,400 hours
Generator	5,000 hours
Control Unit	5,000 hours
Current Transformer	100,000 hours
Maximum Weight	
Generator (dry)	28.65 pounds
Control Unit	2.1 pounds
Current Transformer	0.4 pound
Maximum Speed for Regulation	15,000 RPM
Minimum Efficiency at Rated Load	
Unity Power Factor	85%
0.75 Lagging Power Factor	78%

FIGURE 9. Specification data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

Requirements:

1. Specification: The system shall conform to MIL-G-21480 with the following exceptions to existing paragraphs:

Delete the following paragraphs: 3.4.2.8, 3.4.9.6.1, 3.4.9.6.1.1, 3.4.15, 3.4.15.1, 3.4.16, 3.4.16.1 through 3.4.16.5, 3.5.7.5, 4.4.1.2.1, 4.4.2.2, 4.4.2.3.1, 4.5.20.1, 4.5.20.2, 4.5.22.1 through 4.5.22.3.4, 4.5.23.1 through 4.5.23.1.2, 4.5.24.1 through 4.5.24.3.4, 4.5.25.1 through 4.5.25.2.1, 4.5.28, 4.5.29, 4.5.30, 4.5.32, 4.5.33.2.

3.4.2 Environmental requirements. Delete and add: The system shall meet the requirements of MIL-E-81910 unless otherwise specified by this specification sheet.

3.4.2.1 Temperature and altitude. Delete and add: The system shall meet the temperature-altitude requirements of MIL-E-81910 except no operation required above 30,000 feet. The generator will be tested as a self-cooled component.

3.4.2.2 Humidity. Delete and add: "As required by paragraph 4.5.25 of this specification sheet."

3.4.2.3 Salt spray. Delete and add: "As required by paragraph 4.5.20 of this specification sheet."

3.4.2.4 Fungus. Delete and add: "As required by paragraph 4.5.21 of this specification sheet."

3.4.2.6 Shock. Delete and add: "As required by paragraph 4.5.24 of this specification sheet."

3.4.2.7 Vibration. Delete and add: "As required by paragraph 4.5.23 of this specification sheet."

3.4.3 Operating position. Add: The generator shall operate in positions described in paragraph 4.5.15 of this specification sheet.

3.4.7 Manual system control. Add: "The test function is required."

3.4.8 System capacity. Add: The generating system shall have a rating capacity of 45 KVA and shall conform to the system capacity requirements of 3.4.8 thru 3.4.8.2 of MIL-G-21480 except for the following:

a. Overload and short circuit capacities, and phase displacement and voltage unbalance shall be based on a 30 KVA rating.

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:44Z
Check the source to verify that this is the current version before use.

b. The volt drop in each feeder, at rated load, between the generator terminals and the point of regulation will not exceed 0.5 volts.

c. Parallel operation will not be required.

d. The voltage limits of Figure 4 of this specification sheet pertain.

e. Between 360 Hz and 400 Hz, the generator shall be capable of supplying a load equivalent to 45 KVA at 0.8 lagging power factor, set at 400 Hz, with a minimum phase voltage at 360 Hz, of 105 volts.

f. Between 320 Hz and 400 Hz, the generator shall be capable of supplying a load equivalent to 22.5 KVA at 0.8 lagging power factor, set at 400 Hz, with a minimum phase voltage at 320 Hz of 105 volts, for 5 seconds.

g. The system shall not exceed the rated voltage at 568 Hz with a 5 KVA 1.0 p.f. load connected to the system for 5 seconds.

h. The generator shall be capable of supplying 22.5 KVA at 0.8 power factor for 3 minutes without generator cooling oil.

3.4.8.3 Short circuit capacity. Delete and add: The system shall be capable of producing a minimum of 260 amps single-phase or two-phase line-to-line and line-to-neutral short circuit for 3.5 seconds. The system shall be capable of producing a minimum of 235 amps three-phase line-to-line and line-to-neutral short circuit for 3.5 seconds.

Add: 3.4.9.2 Auxiliary bearing. The generator shall include an auxiliary bearing at the free end. The auxiliary bearing shall not carry any load until the main bearing fails. The generator shall be capable of operating at rated load at least ten hours on the auxiliary bearing after the main bearing fails. The generator shall include a bearing failure warning circuit which shall have a resistance of 5,000 ± 500 ohms to ground before the main bearing fails, and a resistance of not more than one ohm to ground after the main bearing fails. The bearing failure warning circuit shall be tested in accordance with 4.5.35 of this specification sheet.

3.4.9.4 Direction of rotation. Change: "counterclockwise" to "clockwise".

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

3.4.9.6.2 Liquid cooling. Delete and add: The generator shall be cooled by oil conforming to MIL-L-23699. The maximum oil inlet temperature shall be 85 degrees C. At this temperature, the pressure drop across the generator shall be between 29.5 - 33.5 psi at an oil flow of 1.0 gpm, with the generator loaded at 45 KVA and rotating at 12,000 rpm. The oil shall have passed through a 75 micron absolute filter before entering the generator.

Add: 3.4.9.6.2.1 Return screen. The generator shall include a return screen in the oil outlet which shall prevent objects greater than 0.018 inch in diameter from leaving the generator.

Add: 3.4.9.6.2.2 Coolant failure. The generator shall be capable of operating with any rated power factor for the times and electrical loads listed below at 12,000 rpm when tested in accordance with 4.5.34 and 4.5.34.1 of this specification sheet.

45 KVA for 12.5 seconds
30 KVA for 30 seconds
22.5 KVA for 53 seconds
20 KVA for 67 seconds
15 KVA for 120 seconds
10 KVA for 270 seconds
0 KVA for 300 seconds

System performance shall not be degraded.

Add: 3.4.9.6.2.3 External oil leakage. External oil leaks shall not exceed 0.5 cc per hour.

3.4.10.6 Potting compounds. Add: The potting compound in the CT is not required to pass the thermal shock of requirement 47 of MIL-STD-454 but the CT must pass the thermal shock test with the potting compound and remain operational.

3.4.11 Color. Replace: "All system components" with "The generator and generator control unit".

Add: 3.4.12.2 Additional marking. The generator shall be marked to show ROTATION and BOTTOM as shown on Figure 1 of this specification sheet.

3.4.17 Reliability. The system components shall have a minimum observed MTBF (mean-time-between-failure) as defined herein when operated under any combination of input rpm, load and environmental conditions specified herein.

3.5.1 Voltage limits. Change "Figure 3" to "Figure 4 of this specification sheet". Add: Also, the potential shall remain within the limits of Figure 4 of this specification sheet when the generator is operated at overspeed for five seconds with a load drawing 5 KVA at unity power factor.

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:44Z
Check the source to verify that this is the current version before use.

3.5.3 Phase displacement and voltage unbalance. Add: Loads shall be based on a 30 KVA output rating. The maximum unbalance of the phase voltage shall be 1.5, 2.5 and 5 percent respectively for the 1/6, 1/3 and 2/3 load unbalance conditions defined in 4.5.8.1 of MIL-G-21480. The phase angle between phases (excluding harmonics) shall be 120 ± 1.3°, 120 ± 2.3° and 120 ± 4.3° respectively for the 1/6, 1/3 and 2/3 load unbalance condition. For zero and rated balanced three-phase load, no phase output voltage (line-to-neutral) shall differ from the average of the 3-phase voltages by more than 0.5 percent and the phase angles shall be within 120 ± 0.6°.

3.5.5 Integral control power. Replace the last sentence with: "Integral Control Power shall be provided with an external power output for system control in accordance with MIL-G-21480 except as defined in 3.5.5.1 herein.

Add: 3.5.5.1 Power output. The integral control power shall provide 3 amps minimum at a minimum of 19 volts continuous. Contactor pull-in current of 10 amps minimum at a minimum of 17 volts for a minimum of 0.1 seconds shall be provided. This power shall be available over the generator speed range of 9,600 to 12,600 rpm. The integral control power output voltage shall not exceed 30 volts over the generator speed range of 9,600 to 12,600 rpm. The ripple shall not exceed 2 volts peak-to-peak maximum. Output voltage shall not exceed 52 volts at a generator speed of 17,040 rpm for 5 seconds minimum.

Add: 3.5.6.1 System operation. The integral control power shall operate all contactors and indicator lights, both when the system is operating normally and when any part other than the permanent magnet generator or its power conversion section has failed, as long as the generator is being driven within the specified speed range. The system shall be designed so that an external source of 24 ± 8 volts DC may be used to operate the Generator Off warning light if the permanent magnet generator or its power conversion section fails, or the generator is not being driven. However, such power shall not be required for any other operation of the system.

3.5.7.2 Overvoltage. Change in two places "Figure 6" to "Figure 6 of this specification sheet."

3.5.7.3 Undervoltage. Undervoltage protection shall be provided by a static element which functions to disconnect the system from the load bus and deenergize the generator after any phase voltage (line-to-neutral) falls below and remains below 100 ± 5 volts for a period of 3.5 to 5.0 seconds. Manual reset of this protective element shall be possible by putting the generator switch (see paragraph 3.4.7) in the "OFF RESET" position (then to "ON") if the voltage on each phase exceeds 103 ± 2 volts. Reset shall be automatic when the system is shut down. After either automatic or manual reset, the undervoltage protection element shall provide for the system to be automatically reconnected to the load bus within 2 seconds after the lowest phase voltage exceeds 103 ± 2 volts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

3.5.7.4 Underframequency. Underfrequency protection shall be provided by a static element which functions to disconnect the system from the load bus and deenergize the generator within a period of from 1 to 3 seconds after the frequency falls below and remains below 373 ± 3 Hz. After the output frequency exceeds and remains above 377 ± 3 Hz, the underfrequency protection shall automatically reconnect the system to the bus within 2 seconds. There will be a minimum differential of at least 4 Hz between drop-out and pick-up. Provision shall be made in the GCU to disarm the underfrequency protection during flight.

3.5.7.5 Overfrequency. Delete.

3.5.7.6 Feeder fault. Delete and add: The system shall provide for the generators to be deenergized and disconnected from the load bus within 60 milliseconds when the fault current on any phase exceeds any value in the range of 20 ± 5 amps.

3.5.7.8 Overspeed. The generator shall be capable of operating at 17,040 rpm for 5 consecutive seconds without mechanical failure or impairment of subsequent performance. The generator housing shall contain the rotor parts in the event of an overspeed malfunction.

3.5.8 Electromagnetic interference. Delete and add: The system shall be subjected to an electromagnetic interference test in accordance with MIL-E-81910 except that the ten microfarad feedthrough capacitor shall be omitted.

4.4.3.4 Generator feeder voltage drop. Change "5 volt drop" to "0.5 volt drop."

4.5.2 Maximum speed for regulation. Change in third sentence "no load" to "full load at any rated power factor."

4.5.3.1 Qualification tests. Change "Figure 3" to "Figure 4 of this specification sheet."

4.5.3.2 Acceptance test. Change "Figure 3" to "Figure 4 of this specification sheet."

4.5.8.1 Qualification test. Add: "The generator rating is to be considered as 30 KVA for this test."

4.5.12 Electromagnetic interference. Delete and add: The system shall be subjected to an electromagnetic interference test in accordance with MIL-E-81910 except that the ten microfarad feedthrough capacitor shall be omitted.

4.5.15 Operating position. Delete and add: The generator shall be operated for 10 hours at 12,000 rpm in each of the following mounting positions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

a. Anti-drive end 18 degrees down from horizontal with 0 degrees rotation from its mounting position.

b. Anti-drive end 18 degrees down from horizontal with 30 degrees right rotation from its mounting position.

c. Anti-drive end 18 degrees down from horizontal with 30 degrees left rotation from its mounting position.

The generator shall be operated for 15 seconds for each of the following mounting positions.

d. Anti-drive end 90 degrees up from horizontal with 0 degrees rotation from its mounting position.

e. Generator horizontal with 90 degrees right rotation from its mounting position.

f. Generator horizontal with 90 degrees left rotation from its mounting position.

Generator input torques shall be monitored and shall not exceed 325 inch-lbs.

4.5.20 Salt spray. Delete and add: All components shall be subjected to a salt fog test in accordance with MIL-E-81910 except that the duration of the test shall be 24 ± 1 hours. At the completion of this test the system control test of 4.5.3.1(b) of MIL-G-21480, will be conducted with a maximum oil input temperature of 85°C.

4.5.21 Fungus resistance. Delete and add: All components shall be subjected to a fungus test in accordance with MIL-E-81910, with the installation instructions and mounting nuts attached. At the completion of this test the system control test of 4.5.3.1(b) of MIL-G-21480, will be conducted with a maximum oil input temperature of 85°C. The mounting nuts shall be unaffected and the package holding the nuts shall be suitable for shipment. Instructions shall be legible. Verification by analysis is allowed if all materials are proven to be non-nutrient.

4.5.22 Sand and dust. Delete and add: "Conduct test as per MIL-E-81910 except that air velocity in steps 1, 2, and 3 shall be 200 ± 100 fpm.

4.5.23 Vibration. Add: All system components shall be vibrated in accordance with Procedure I, Method 514.2, MIL-STD-810. Procedure I shall be accomplished using the vibration test curve of Figure 5 of this specification sheet and the test time per axis will be 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

hours less the time spent in resonant dwells. Any major resonances up to a total of three shall be performed for 30 minutes each. Sinusoidal cycling shall not be conducted for more than 180 minutes or less than 60 minutes per axis. The sweep time for sinusoidal cycling over the frequency range of 5 to 500 Hz and back to 5 Hz shall be 15 minutes. The component shall be operated at no load, but other applicable conditions such as rated voltage, frequency, and speed shall be maintained. At the conclusion of the test, the component shall be operated and inspected. There shall be no evidence of cracking or rupture of any part of the component or of the mounting assembly. The item is considered failed if during or after any test herein, the item will not meet the specified performance limits, or if the maintenance of the item is impeded, or if the item requires the replacement of components.

4.5.23.2 Acceptance test, vibration. Delete and add: Control components shall be subjected to the burn-in test in accordance with 4.6.10 of MIL-E-81910.

4.5.24 Shock. Delete and add: All components shall be subjected to a shock test in accordance with MIL-E-81910. At the completion of Procedure I, the system control test of 4.5.3.1(b) of MIL-G-21480, will be conducted with a maximum oil input temperature of 85°C. There shall be no failure of the mounting attachments, and the test item shall remain in place and not create a hazard during Procedure III.

4.5.25 Humidity. Delete and add: All components shall be subjected to a humidity test in accordance with MIL-E-81910. At the completion of this test the system control test of 4.5.3.1(b) of MIL-G-21480, shall be conducted.

4.5.33 Reliability.

4.5.33.1 Reliability test. Delete and add: "The reliability demonstration test shall be structured as a test, analyze and fix (TAAF) program wherein any failure (any operation outside of specified limits) which occurs is analyzed and corrective action implemented by the contractor to prevent recurrence of the failure. For the generator and GCU a minimum of 5000 hours total operating time on two samples shall be accumulated with one or less failures of each component under the test conditions of the endurance test of 4.5.14 in order to successfully complete the test. In the event more than one failure of either component occurs, additional operating time shall be accumulated until the total operating hours divided by the number of failures of each component equals 5000 hours or higher. When a corrective action has resolved a failure to the satisfaction of the qualifying activity, all but the first occurrence of the resolved failure shall be dropped from consideration. No single generator or GCU of the two test samples shall accumulate less than 25 percent of the total test time. During the test, the output voltage shall remain within the limits of Figure 4. Following the test, the system shall pass the system control test of 4.5.3.1(b) of MIL-G-21480."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:44Z
Check the source to verify that this is the current version before use.

Add: 4.5.34 Failure of oil cooling supply. With the generator temperature stabilized, the generator shall be operated at the loads and times listed in 3.4.9.6.2.2 of this specification sheet with the oil supply shut off. In addition, the test under paragraph 3.4.8(h) shall be conducted.

Add: 4.5.34.1 External oil leakage. During all of the tests specified herein, the external oil leakage from the generator shall not exceed the limit of 3.4.9.6.2.3.

Add: 4.5.35 Auxiliary bearing. A test shall be conducted to demonstrate compliance with paragraph 3.4.9.2 as follows:

Contaminate the main bearing to cause failure. Operate on the auxiliary bearing at 12,000 rpm, 45 KVA 0.8 p.f. for 10 hours.

The bearing failure warning light shall remain illuminated throughout the 10 hour test period.

5.1 Preservation, packaging, and packing. Delete and add: The acquiring activity will specify the applicable levels of packaging and packing.

6.2.2 Point of regulation. In last sentence change "five" to "0.5".

6.4 Qualification. Delete in line 8 "(Code WST 33)" and replace with "(Code SY60)".

Preparing Activity:
Navy - AS
(Project No. 6115-N484)

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:44Z
Check the source to verify that this is the currentversion before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department
(ESSD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON, D.C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(ESSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:44Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-21480/23	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-19T09:44Z
Check the source to verify that this is the current version before use.